Assignment No.5 2413FEB2T1 : Engineering Mathematics-2
Topic: Beta and Gamma Functions, DUIS

1 Problems on Gamma Function

1. Evaluate $\int_0^\infty y^6 e^{-y^6} dy$
2. Prove that $\int_0^\infty \sqrt{y} e^{-y^2} dy \int_0^\infty \frac{e^{-y^2}}{\sqrt{y}} dy = \frac{\pi}{2\sqrt{2}}$
3. Show that $\int_0^\infty x e^{-x^8} dx \int_0^\infty x^2 e^{-x^4} dx = \frac{\pi}{16\sqrt{2}}$
4. Evaluate $\int_0^\infty e^{-x^5} dx$
5. Evaluate $\int_0^\infty 7^{-4x^2} dx$
6. Evaluate $\int_0^\infty x^n e^{\sqrt{ax}} dx$

2 Problems on Beta Function

1. Evaluate $\int_0^2 x^4 (8 - x^3)^{-\frac{1}{3}} dx$
2. Evaluate $\int_0^1 \sqrt{\sqrt{x} - x} dx$
3. Evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{sin\theta}} \int_0^{\frac{\pi}{2}} \sqrt{sin\theta} d\theta$
4. Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{cot\theta} d\theta$
5. Evaluate $\int_0^{\frac{\pi}{6}} cos^6(3\theta) sin^2(6\theta) d\theta$
6. Evaluate $\int_0^\infty \left(\frac{x}{1+x^2}\right)^4 dx$
7. Evaluate $\int_0^\infty \frac{x^8(1-x^6)}{(1+x)^{24}} dx$
8. Evaluate $\int_3^7 \sqrt[4]{(x-3)(7-x)} dx$

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9. Evaluate $\int_0^4 \sqrt{x}(4-x)^{\frac{3}{2}}dx$

10. Evaluate $\int_0^\infty \frac{x^4(1-x^5)}{(1+x)^{15}}dx$

11. If $\beta(n,3) = \frac{1}{105}$ and n is a positive integer then find the value of n.

3 Problems on Differentiation Under Integral Sign(DUIS)

1. Show that $\int_0^\infty \frac{tan^{-1}(ax)}{x(1+x^2)}dx = \frac{\pi}{2}log(1+a)$

2. Evaluate $\int_0^1 \frac{x^a-1}{logx}dx$

3. Prove that $\int_0^\infty \frac{log(1+ax^2)}{x^2}dx = \pi\sqrt{a}$

4. Evaluate $\int_0^\pi \frac{log(1+acosx)}{cosx}dx = \pi sin^{-1}a$ for $0<a<1$.

5. Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1+acos^2x}$ and hence deduce that $\int_0^{\frac{\pi}{2}} \frac{cos^2x}{(3+cos^2x)^2}dx = \frac{\pi\sqrt{3}}{96}$.

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